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Experimental Investigation of flint-based porous media cooling for enhanced photovoltaic performance

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ABSTRACT

Photovoltaic (PV) panels undergo significant efficiency losses and reduced reliability due to temperature rise during operation. This study introduces a hybrid cooling system combining flint-based porous media with water-cooling channels to enhance heat dissipation. Flint, known for its high thermal conductivity and durability, is employed as a natural porous material to improve conductive and convective cooling. Outdoor experiments were conducted on 30 W polycrystalline PV modules to evaluate the influence of porosity (0.35–0.48) and coolant flow rate (1–2 L/min) on thermal performance. A three-stage protocol was followed: (1) comparing porous versus non-porous cooling, (2) determining the optimal porosity, and (3) calibrating flow rate for peak efficiency. Results showed a 27% temperature reduction, with surface temperatures dropping from 50.9°C to 37.7°C under peak conditions. The system improved power output by 12.3% and electrical efficiency by 13.5%, with optimal performance at 0.35 porosity and 2 L/min. The hybrid design balanced cost and efficiency, achieving a lower levelized cost of energy ($0.103/kWh) than water cooling alone ($0.105/kWh). Environmental analysis revealed a 360.15 kg CO_2 reduction over 15 years and a 3.5% annual efficiency gain compared to uncooled PV. Although annual carbon credit gains were modest ($0.36), the system's scalability offers long-term sustainability benefits.

KEYWORDS

Flint porous media; heat dissipation; photovoltaic performance; porosity; PV cooling

1. Introduction

The operating temperature of photovoltaic (PV) panels represents a fundamental limiting factor in solar energy conversion efficiency and system longevity (Dev, Ravi, and Saini 2022). As semiconductor devices, PV cells exhibit strong temperature-dependent performance characteristics governed by fundamental physics principles. Elevated temperatures induce two primary degradation mechanisms: (1) reduction of open-circuit voltage (Voc) due to increased carrier recombination rates and (2) alteration of the semiconductor bandgap energy (Gomaa et al. 2020). El Kassar et al. (2023) reported a significant efficiency reduction in PV cells, from 21.3% at 25°C to 17.41% at 70°C, corresponding to an average temperature coefficient of approximately −0.4% to −0.5% per °C for crystalline silicon technologies. Similarly, Al-Ghezi, Ahmed, and Chaichan (2022) found that each 1°C increase in temperature led to an efficiency loss of approximately 0.586%. Ale, Rotipin, and Makanju (2022) emphasized the strong correlation between PV temperature, short-circuit current, solar irradiance, and efficiency, highlighting the need for effective temperature control. The thermal challenge is further compounded by real-world operating conditions where PV modules frequently exceed 20–30°C above ambient temperature due to absorbed but unconverted solar radiation. Field studies by Armstrong and Hurley (2010) and Kempe et al. (2021) reveal that under peak insolation, module temperatures can reach 60–80°C depending on mounting configuration and environmental factors. These elevated temperatures not only reduce instantaneous power output but also accelerate long-term degradation processes, including encapsulant discoloration, solder bond fatigue, and potential-induced degradation, collectively reducing system lifespan by 2–5 years under typical operating conditions (Al Mahdi et al. 2024). Skoczek et al. (2007) identified three primary temperature-induced failure modes: encapsulant discoloration, solder bond

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fatigue and interconnect failure, and potential-induced degradation. As temperatures increase, the Voc exhibits an almost linear decrease over a broad temperature range, while the short-circuit current tends to increase (Bahaidarah, Baloch, and Gandhidasan 2016). These thermally induced changes result in a reduction in the system's overall power output, highlighting the critical need for effective thermal management in PV systems (Ali et al. 2024). Without adequate cooling or heat dissipation, PV panels not only experience reduced power generation but may also face a shortened operational lifespan, ultimately driving up maintenance and replacement costs. Consequently, optimizing thermal management is critical to improving both the efficiency and durability of PV systems under diverse environmental stresses (Al-Dahidi et al. 2025). Implementing effective thermal management strategies can minimize excessive heating, mitigate efficiency losses, and maximize the long-term benefits of solar technology. As highlighted by Shaker et al. (2024), addressing temperature-induced performance changes necessitates the adoption of efficient cooling techniques, such as active or passive cooling systems, to improve both the efficiency and durability of PV panels (Hachim et al. 2025).

To address these challenges, significant research has been dedicated to developing advanced cooling techniques for PV panels (Attia et al. 2024), including active and passive cooling systems, phase change materials (PCMs) (Ali et al. 2025), nanofluids, and hybrid cooling strategies (Abdul Rahim et al. 2024). PCMs have shown considerable promise in regulating PV temperatures (Alahmer et al. 2025). For example, Jakhar, Soni, and Gakkhar (2017) integrated a heat exchanger with a concentrating PV system, reducing temperatures from 416.36°C to 85.28°C. Elavarasan et al. (Madurai Elavarasan et al. 2024) integrated PCM, fins, and water for PV cooling, achieving a maximum power conversion efficiency of 20.13%, significantly higher than uncooled systems. Čorić et al. (2024) highlighted the effectiveness of PCM-based cooling, showing a 13.93% increase in output power and a 17.93% reduction in temperature using latent heat storage units. Farhan, Naqvi, and Uzair (2024) confirmed PCM integration as a viable passive cooling solution, reducing PV panel temperatures by 18°C and increasing electrical power by 10.89%.

Nanofluids have also been widely studied for their cooling potential. Abdallah et al. (2018) demonstrated that using aluminum oxide (Al_2O_3)-water nanofluid in PV/thermal (PV/T) systems improved performance by 56% at an optimal flow rate of 1.2 L/min. Al-Waeli et al. (2017) reported enhanced cooling and energy output using silicon carbide (SiC) nanofluid. Ebaid, Ghrair, and Al-Busoul (2018) found that Al_2O_3 nanofluid outperformed titanium dioxide (TiO_2) in reducing PV temperatures. Elminshawy et al. (2021) demonstrated that nanofluids in low-concentration PV systems improved electrical efficiency and cooling performance.

Hybrid cooling strategies, which combine multiple cooling techniques, have proven particularly effective. Peng, Herfatmanesh, and Liu (2017) designed an optimized cooling model for solar PV panels, achieving a total system energy efficiency of up to 107% and a 35% increase in electric output efficiency. Mahdi et al. (2024) enhanced PV/T systems using a hybrid solar collector with PCM, porous media, and nanofluids, increasing daily exergy and thermal energy to 14.32% and 302.72 W, respectively. Saeed et al. (2024) showed that combining PCM with fins reduced PV temperatures by 11–15% and improved efficiency by 2.1%. Micheli, Reddy, and Mallick (2015) demonstrated the potential of plate micro-fins for passive cooling in concentrated PV systems, while Mojumder et al. (2016) achieved a PV efficiency of 13.75% and thermal efficiency of 56.19% using air-type PV/T collectors with cooling fins. Moradgholi, Nowee, and Abrishamchi (2014) improved heat transfer in a novel PV/T system, increasing thermal efficiency from 16.35% to 45.14%.

Water-based cooling systems have also been widely explored. Smith et al. (2014) proved the effectiveness of water cooling in large-scale applications, maintaining PV temperatures below 40°C and enhancing energy efficiency. Syafiqah et al. (2017) showed that water cooling outperforms air cooling, reducing PV temperatures by 53.2%. Tang, Hua Quan, and Hua Zhao (2010) and Wu, Chen, and Xiao (2018) highlighted the potential of micro heat pipe arrays and water-cooled PV/T systems in improving thermal performance and energy conversion efficiency.

Innovative cooling methods, such as pulsating heat pipes (PHP), evaporative cooling, and porous media, have further advanced PV thermal management. Alizadeh et al. (2018) found that PHP systems increased electrical power generation by 18%, while Alami (Alami 2014) highlighted the effectiveness of evaporative cooling in hot climates. Golzari et al. (2018) demonstrated significant heat dissipation improvements using corona wind. Castanheira, Fernandes, and Branco (2018) implemented large-scale cooling systems in Portugal, increasing annual energy production by 12%.

Porous materials have emerged as a highly effective solution for thermal management in PV systems, offering unique advantages in heat dissipation and temperature regulation. Their inherent structure, characterized by interconnected voids and high surface area, facilitates efficient heat transfer and uniform cooling across PV panels (Mansour et al. 2023). Recent studies have demonstrated the potential of porous materials to significantly improve PV efficiency and power output by maintaining optimal operating temperatures. Masalha et al. (2025) developed a porous media – integrated cooling system, combining experiments and ANSYS simulations, and optimized porosity, flow rate, coolant type, and channel design, achieving up to 35.7% temperature reduction and 9.4% power gain. Building on this, Masalha, Badran, and Alahmer (2025) investigated low-cost natural porous materials (sandstone, flint, marble, and gravel) for 30 W PV panels under controlled solar conditions, finding that sandstone with 0.35 porosity and 2 L/min water flow reduced panel temperature by 58%, increased efficiency by 57.9%, and boosted power output by 57.8%. The study highlighted that cooling effectiveness strongly depends on material thermal conductivity and optimized flow and porosity parameters.

The integration of porous materials with other cooling techniques, such as nanofluids, has also shown remarkable results. Al-Amir et al. (2023) studied TiO_2-water nanofluid in a porous cavity, revealing an 80% increase in heat transfer as the Rayleigh number (Ra) increased from 10^5 to 10^6. Al-Dulaimi et al. (2024) demonstrated the superior heat transfer performance of silver-magnesium oxide (Ag-MgO) hybrid nanofluids in corrugated porous enclosures, further highlighting the synergistic effects of combining porous materials with advanced fluids. These hybrid approaches not only improve heat dissipation but also enhance the overall energy efficiency of PV systems.

The effectiveness of porous materials is further supported by their ability to maintain uniform cooling across PV panels. Bahaidarah, Baloch, and Gandhidasan (2016) emphasized the importance of uniform cooling strategies in reducing temperature-related efficiency losses, a challenge that porous materials are well suited to address. Additionally, Sun et al. (Sun et al. 2017) explored the thermal conductivity of flint, a material relevant to porous cooling systems, revealing a linear relationship between thermal conductivity and saturation levels above 15%. This study highlights the importance of material properties in designing effective porous cooling solutions.

Various porous materials have been investigated for PV cooling systems, each presenting distinctive benefits and limitations. Gravel has been widely utilized due to its ready availability (Masalha et al. 2025), yet its relatively low hardness (H = 3–4) renders it susceptible to erosion under continuous water flow, resulting in structural deterioration and compromised heat transfer efficiency. Furthermore, its modest thermal conductivity (k ≈1.19 W/m·K) restricts its overall cooling performance. Glass has found application in thermal management systems (Ahmed, Hamada, and Salih 2019), but its inherent fragility makes it vulnerable to microfractures during prolonged water exposure, while its thermal conductivity (k ≈0.8 W/m·K) proves inadequate for optimal heat dissipation. Plastic materials, despite their corrosion resistance, demonstrate severely limited effectiveness due to their extremely low thermal conductivity (k ≈0.243 W/m·K) (Salman, Hilal, and Ghadhban 2022), compounded by UV-induced degradation and potential chemical leaching issues. In contrast, flint emerges as a superior alternative, boasting exceptional hardness (H = 8), remarkable erosion resistance, and outstanding thermal conductivity (k ≈3.5 W/m·K). The material's textured surface promotes even water distribution, effectively minimizing localized hot spots and significantly enhancing cooling performance. These superior characteristics position flint as the most promising porous medium for sustainable PV cooling applications, outperforming conventional alternatives.

As summarized in Table 1, previous research has investigated a range of PV cooling methods, including PCMs, nanofluids, and water-based systems, each presenting limitations such as material instability, high cost, or limited real-world applicability. Although porous materials have been explored for thermal management, the integration with conventional cooling techniques remains underdeveloped, particularly in practical outdoor scenarios. Remarkably, no prior studies have focused on flint as a porous medium, despite its advantageous properties.

This study presents a novel hybrid cooling system that integrates flint-based porous media with forced water flow to enhance both conductive and convective heat transfer in PV panels. Flint's high thermal conductivity (~3.5 W/m·K), exceptional hardness (H = 8), and resistance to corrosion make it a durable and efficient alternative to conventional porous materials. In contrast to gravel, glass, and plastic – which suffer

Table 1. Comparative summary of PV cooling studies and performance.

Study	Cooling Method	Medium Used	Key Parameters	Performance Enhancement	Key Findings/Limitations
(Jakhar, Soni, and Gakkhar 2017)	Earth Water Heat Exchanger (EWHE)	Water	Mass flow rate 0.022 kg/s; CPV at 3 Suns	Temp drop from 416.36°C to 85.28°C; power output increased with flow rate	Effective cooling via EWHE; simulation based, no real-world validation
(Madurai Elavarasan et al. 2024)	PCM + Fins + Still Water (Passive)	HS 29 PCM, fins, water	Location-adapted PCM; latent heat utilization	Max temp drop 16.7°C; power up to 20.25% higher; avg 8.57% efficiency gain	Passive cooling effective; corrosion and thermal stress analyzed; climate-specific
(Čorić et al. 2024)	Review of PCM passive cooling	Paraffin PCM + fins/ additives	Various PCM setups: single, dual, additives, fins, radiative cooling	Up to 17.93% temp reduction; 13.93% power increase (LHSU with fins)	Extensive review; noted gaps in long-term field data and durability
(Farhan, Naqvi, and Uzair 2024)	PCM Backside Cooling	Soy wax PCM	Rear panel cooling; hybrid system	Temp drop up to 18°C; power increased 10.89%	Low-cost, effective; short-term results; durability not assessed
(Abdallah et al. 2018)	Nanofluid cooling in PV/T system	Al_2O_3 nanoparticles in water	Nano fluid concentrations 0.05–0.3 vol %; flow rate optimum at 1.2 L/min	Combined PV/T performance increased by 74% (thermal) and 56% (electrical) at 0.1 vol%	Nano fluid significantly boosts PV/T efficiency; flow rate optimization important; experimental study
(Al-Waeli et al. 2017)	Nanofluid cooling in PV/T system	SiC nanoparticles in water (3 wt%)	Thermal conductivity, viscosity, density measured over 25–60°C	Electrical efficiency increased by 24.1%; thermal efficiency increased by 100.19%	Nanofluid stable over 6 months; suitable for long-term use; total system efficiency ~ 88.9%
(Ebaid, Ghrair, and Al-Busoul 2018)	Nanofluid cooling with heat exchanger	Al_2O_3 and TiO_2 nanofluids in water-polymer mixtures	Nanofluid conc. 0.01–0.1 wt%; flow rates 0.5–5 L/min; pH levels	Al_2O_3 nanofluid lowered PV temp more; TiO_2 nanofluid showed better electrical performance	Higher conc. improves cooling; Al_2O_3 better temp reduction; TiO_2 better power output; outdoor tests in Jordan
(Elminshawy et al. 2021)	Nanofluid cooling for low concentration PV (LCPV)	Al_2O_3/water nanofluid at 1–3 vol%	Volume fraction ratio effects; operating temp < 45°C targeted	Power output improved 13.58% (water cooling), 24.02% (3% nanofluid) vs uncooled; temp dropped by 16.47°C	Higher nanofluid concentrations reduce temperature significantly; strong enhancement in electrical and thermal efficiency
(Peng, Herfatmanesh, and Liu 2017)	Water-based active cooling system	Water	Radiation conditions; lifecycle assessment	Instantaneous efficiency up to 47%; annual electric output + 35%	System payback reduced to 12.1 years; large gains in hybrid output
(Mahdi et al. 2024)	Hybrid PVT-HAWSC with PCM, nanofluid	MWCNT-water nanofluid, paraffin PCM, steel wool porous media	Air/water flow rate variations; double-pass duct	PV temp drop 28°C (vs 22°C without nanofluid); electrical eff. 93.64%	Highly efficient system; reduced LCOE; experimental validation
(Saeed et al. 2024)	Passive cooling: PCM vs fins	PCM, aluminum fins	Back panel attachment; real outdoor tests	Temp drop 11–15% (PCM), 7–13% (fins); 2.1% and 1.7% power gain	PCM superior to fins; practical, low-cost passive methods
(Micheli, Reddy, and Mallick 2015)	Micro-fins for CPV cooling	Aluminum micro-fins	Fin geometry optimization	Mass specific power improved up to 50%	Lightweight and thermally effective; limited published studies
(Mojumder et al. 2016)	Air-cooled PV/T with fins	Air, thin flat metallic fins	Number of fins (0–4), flow rates 0.02–0.14 kg/s	Thermal efficiency up to 56.19%, PV efficiency 13.75%	Experimental validation with correlations; flow and solar radiation effects studied
(Moradgholi, Nowee, and Abrishamchi 2014)	Heat pipe (thermosyphon) PV/T system	Methanol, acetone	Seasonal angle adjustments (30°, 40°)	Electrical power + 5.67 to + 7.7%; thermal efficiency + 16–45%	Effective temp reduction (up to 15°C); seasonal performance variation

(*Continued*)

Table 1. (Continued).

Study	Cooling Method	Medium Used	Key Parameters	Performance Enhancement	Key Findings/Limitations
(Syafiqah et al. 2017)	Dual cooling: air (back) + water (front)	Water and air	ANSYS CFX and PSPICE modeling	Temp reduction 53.2% (water), 19.2% (air)	Water cooling more effective; simulation study needs experimental data
(Alizadeh et al. 2018)	Pulsating Heat Pipe (PHP) cooling	PHP vs copper fin	Numerical simulation	~18% electrical power increase using PHP	PHP promising for PV cooling; requires experimental confirmation
(Alami 2014)	Passive evaporative cooling	Synthetic clay + thin water film	Indoor experimental setup	Output voltage + 19.4%, power + 19.1%	Simple, silent, cost-effective; suitable for arid climates
(Golzari et al. 2018)	Electro-hydrodynamics (EHD) cooling	Air with corona wind	Voltage up to 11 kV; air flow rate variation	Heat transfer coeff. +65%; thermal efficiency + 28.9%	EHD effectively enhances heat transfer; electrical input considerations
(Castanheira, Fernandes, and Branco 2018)	Water cooling kit for existing PV plants	Water	20 kW plant rooftop tests	Annual energy gain up to + 12%	Effective retrofit cooling; water & energy loss optimized; economic analysis done
(Masalha et al. 2025)	Porous media – integrated cooling system	Water, chemical alcohol, engine oil	Porosity size (0.35–0.5), flow rate (1–4 L/min), cooling channel design, porous media type (gravel, marble, flint, sandstone) (Based on ANSYS simulations)	Up to 35.7% PV temperature reduction, 9.4% power output increase	Optimized system achieved best performance with porosity 0.35, flow rate 2 L/min, tapered channel (5 mm inlet –3 mm outlet), and sandstone medium. ANSYS simulations validated experimental results; study focuses on 30 W polycrystalline PV cells.
(Masalha, Badran, and Alahmer 2025)	Porous media cooling	Water	Porosity (0.35–0.48), flow rate (1–4 L/min), porous material type (sandstone, flint, marble, gravel) (Indoor testing under controlled laboratory conditions)	Up to 58% PV temperature reduction	Sandstone with 0.35 porosity and 2 L/min flow rate was most effective; cooling performance strongly depends on material thermal conductivity and optimized flow/porosity.
Current Study	Hybrid porous media-liquid cooling	Flint-based porous media + water	Porosity (0.35–0.48); flow rate (1–2 L/min); 30 W polycrystalline PV modules (Outdoor, real situation)	Temperature reduced by 27% (from 50.9°C to 37.7 °C); power output ↑ 12.3%; electrical efficiency ↑ 13.5%	Economic analysis shows LCOE $0.103/kWh, better than water-only ($0.105/kWh); 360.15 kg CO_2 avoided over 15 years.

from low thermal performance and long-term degradation, flint offers superior stability and thermal efficiency. Despite its potential, flint remains underexplored in PV cooling applications, and there is limited experimental data on how pore structure and fluid dynamics interact to influence system performance under real outdoor conditions. This study fills that gap by experimentally investigating the combined thermal effects of flint-based porous media and water flow, offering valuable insights into improved thermal management strategies for photovoltaic systems.

This study aims to systematically evaluate the thermal performance of a hybrid flint porous media-water cooling system for PV modules through experimental analysis. The research establishes five key objectives:

- First, to quantify the cooling performance of flint-enhanced porous cooling against conventional liquid-only cooling and uncooled PV systems.
- Second, to analyze the influence of porosity characteristics (ranging from 0.35 to 0.48) and coolant flow rate (1–2 L/min) on heat dissipation and electrical efficiency.
- Third, to validate these findings through real-world testing by conducting experimental testing on 30 W polycrystalline solar modules to assess temperature reduction, power output enhancement, and operational stability under practical environmental conditions.
- Fourth, to investigate the combined effects of conduction (through porous media) and forced convection (via liquid cooling) to develop an optimized thermal regulation strategy.
- Fifth, to evaluate the economic feasibility and environmental impact of the proposed cooling system.

This research introduces several key innovations in PV cooling technology:

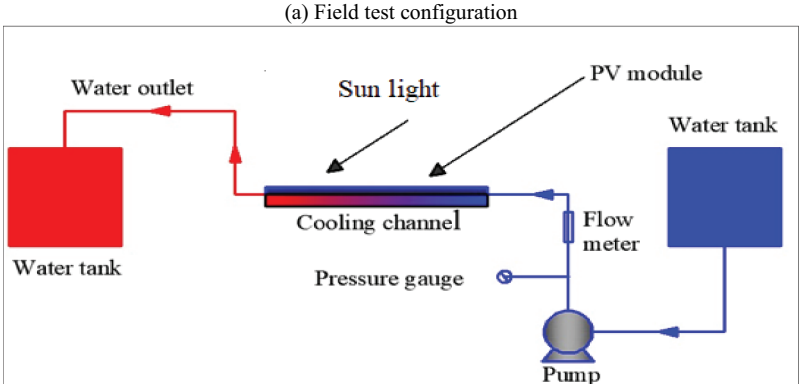
- It introduces a fundamentally new approach to PV cooling by integrating flint-based porous media with water cooling channels, a combination not previously explored in PV thermal management. Flint is employed as a natural porous medium, offering a sustainable, cost-effective, and durable alternative to synthetic materials. Flint's high thermal conductivity, hardness, and resistance to corrosion contribute to improved heat dissipation and long-term operational stability without the need for frequent maintenance.
- A unique hybrid cooling configuration is developed, combining heat conduction through porous media with forced liquid convection. This synergy results in more effective thermal regulation than conventional single method cooling systems.
- This study is among the first to apply a Monte Carlo simulation with 1,000 iterations to account for ± 20% variability in annual energy output due to operational and environmental uncertainties. This probabilistic approach provides a more realistic and robust estimation of the LCOE for PV systems with and without cooling, improving the reliability of cost-performance comparisons.
- This study is among the first to comprehensively evaluate the environmental performance of a small-scale (30 W) PV system under different cooling configurations by integrating annual energy output, net CO_2 mitigation, embodied emissions, and carbon credit valuation.
- The irregular surface morphology and pore structure of flint enhance fluid interaction within the cooling channel, boosting both conductive and convective heat transfer. The flow disruption caused by the porous matrix also induces micro-vortex turbulence, which further improves thermal exchange.
- Unlike previous studies that rely solely on laboratory simulations, this research evaluates the cooling system under variable flow rates (1–2 L/min) and real-world environmental conditions. This provides more practical insights for the implementation of PV cooling systems in actual operating scenarios.
- Optimal porosity and flow rate ranges are identified to maximize cooling effectiveness while minimizing energy consumption, offering a scalable and energy-efficient solution for improving PV module performance.

2. Research methodology

A 5-cm-thick cooling channel was attached to the back of a solar panel to study how cooling influences PV performance. The channel was meticulously fabricated to match the panel's dimensions, ensuring uniform heat dissipation. The cooling system utilized locally sourced crushed flint as the porous medium, selected for



(a) Field test configuration



(b) Functional diagram illustration

Figure 1. (a) Field experimental apparatus and (b) technical schematic of the PV panel cooling system during outdoor evaluation.

its favorable thermal conductivity and porosity. The flint was tested at three different porosity levels (0.35, 0.40, and 0.48) while water was circulated at varying flow rates (1, 1.5, and 2 L/min) over a 7-hour testing period.

As illustrated in Figure 1, the experimental setup consisted of PV panels for electricity generation, combined with a water-based cooling system to manage thermal output. The cooling channels played a vital role in dissipating excess heat by circulating water along the back surface of the panels. This process effectively absorbed and removed thermal energy, maintaining a lower operating temperature and enhancing the system's overall efficiency.

The experimental study was designed to assess the performance of different cooling configurations through three distinct scenarios.

In the first scenario, the study compared the cooling effectiveness of a porous medium with a porosity of 0.35 against two alternative setups: a system utilizing only water without any porous media and a control system without any cooling mechanism. This comparative analysis was designed to elucidate the role of porous media in enhancing thermal dissipation within PV systems.

The second scenario expanded the investigation by testing three cooling channels filled with porous materials of varying porosities (0.35, 0.40, and 0.48) alongside the uncooled system. This phase aimed to evaluate how different pore structures affect the thermal regulation of PV panels.

In the third scenario, the focus shifted to the effect of water flow rate on cooling efficiency. Three cooling channels were filled with flint porous media of a constant porosity (0.35) while varying the water flow rates. This setup allowed for an in-depth analysis of how flow rate adjustments influence heat dissipation and PV panel performance.

2.1. Test design and measurements

The test setup, as displayed in Figure 1, was mounted on the rooftop of the Faculty of Engineering Technology at Al-Balqa Applied University. To maintain experimental consistency, all PV panels were

evaluated simultaneously under uniform conditions. While solar irradiance and ambient temperature exhibited natural variations throughout the day, data collection was strategically timed during stable environmental periods. As illustrated in Figure 1(a), three polycrystalline silicon PV panels were used, each fitted with rear-surface cooling channels – except for one panel, which served as an uncooled baseline for comparison. Cooling fluid was circulated using two water pumps connected to a 1 m^3 storage tank via a polyvinyl chloride (PVC) piping network. To monitor temperature variations across the PV panels, K-type thermocouples were strategically positioned at multiple locations on each panel's surface. Figure 1(b) offers a detailed view of the PV cells and cooling channels. Additionally, two flow meters were installed at the pump outlets to continuously monitor and regulate the cooling fluid's flow rates. To further enhance the dataset, various other instruments were employed to measure solar radiation intensity, as well as the inflow and outflow temperatures of the water-cooling fluid.

2.2. Cooling system design

The experimental hybrid PV/T system comprised monocrystalline silicon PV panels, each with a nominal power output of 30 W under standard test conditions (STC). To enhance thermal regulation and mitigate efficiency losses due to overheating, an aluminum-finned thermal collector was integrated onto the rear side of each PV panel. The collector was designed to match the exact dimensions of the panel's rear surface, ensuring uniform heat dissipation and optimal cooling coverage. A visual representation of this configuration, illustrating the effective integration of the cooling system, is provided in Figure 2, while Table 2 presents detailed technical specifications of the PV panels, including their electrical characteristics, temperature coefficients, and operational thresholds.

The cooling system employed a closed-loop water circulation mechanism, with strategically designed inlet and outlet manifolds to ensure a laminar and consistent flow of coolant across the panel surfaces. The heated water exiting the cooling channels was channeled into an insulated storage tank, effectively preventing thermal mixing with the incoming chilled water supply. This segregation was critical for maintaining thermodynamic efficiency, as any recirculation of heated coolant would reduce the cooling capacity and impair PV performance. A 0.37 kW centrifugal pump, chosen for its energy efficiency and reliability, circulated the coolant through a network of high-durability PVC piping, ensuring low thermal resistance and consistent heat exchange performance.

A cooling channel was installed beneath each PV panel to regulate temperature and improve efficiency, as shown in Figure 2. Two configurations were experimentally tested: (1) water-only cooling and (2) water cooling enhanced with porous media. In the water-only configuration, the cooling channel beneath the panel was left empty, allowing water to flow directly from the inlet to the outlet. As the water moves along the base of the PV panel, it absorbs heat through forced convection, reducing the temperature of the panel surface. This method relies primarily on the water's specific heat capacity and flow velocity for effective thermal regulation. However, due to the limited contact surface area between the water and the panel, the rate of heat transfer is constrained. In the porous media-enhanced configuration, the same cooling channel was filled with flint, a naturally occurring, thermally stable, and highly porous material. This porous medium creates a network of interconnected voids and solid surfaces, which significantly increases the contact area between the flowing coolant and the medium. As water passes through this structure, it is subjected to enhanced conduction through the solid flint and evaporation from the extended surface area exposed to airflow, resulting in a dual-mode heat transfer mechanism. Additionally, the irregular structure of the porous media disrupts the flow, promoting turbulence at the microscale, which further improves heat exchange efficiency. These differences in configuration and heat transfer behavior explain the improved thermal performance of the porous media-enhanced cooling system.

The experimental system incorporated a comprehensive array of calibrated instrumentation to monitor all critical parameters in real time. As illustrated in Figure 3, the setup included the following: (1) a digital multimeter for measuring PV panel electrical output (voltage, current, and resistance); (2) transparent-tube flow meters with graduated scales to precisely determine water flow rates; (3) an electronic temperature data logger that recorded and graphically displayed thermal variations across the PV surface and cooling channels; (4) a digital pyranometer with sensor for measuring solar radiation intensity; and (5) an electric water pump that maintained controlled circulation through the piping network. The data acquisition

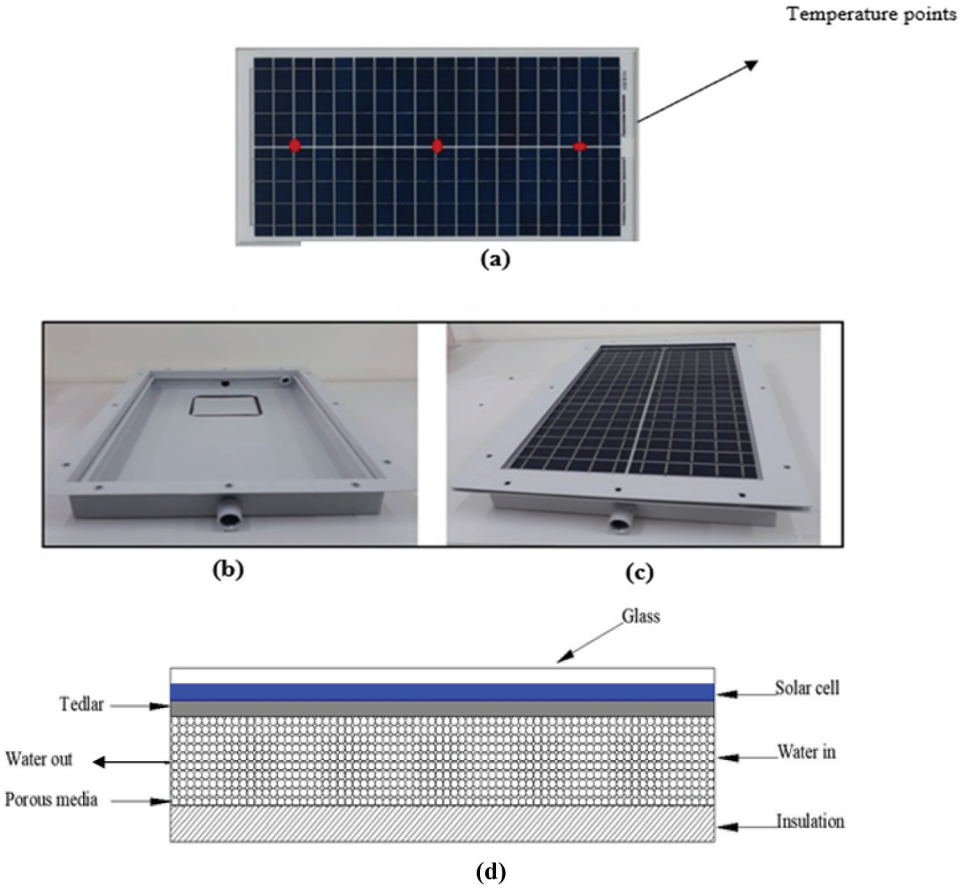


Figure 2. Integrated cooling system for PV modules showing: (a) standalone PV, (b) isolated cooling channel assembly, (c) combined PV-cooling channel unit, and (d) schematic representation of the complete cooled PV system.

Table 2. PV panel specifications and characteristics.

Solar PV module parameter	Value @ standard test condition
Maximum power	30 W
Open-circuit voltage	22 V
Short-circuit current	1.84 A
Voltage at maximum power	18 V
Current at maximum power	1.67 A
Operating temperature range	−40°C to+85°C
Solar intensity	1000 W/m^2

system captured these parameters – including panel temperatures, coolant flow rates, irradiance levels, and power output – at regular intervals, enabling precise quantification of cooling performance under varying operating conditions. This integrated monitoring approach ensured reliable data collection for evaluating the hybrid cooling system's effectiveness while maintaining controlled fluid dynamics and thermal regulation.

Electrical power output was measured using a digital multimeter, which recorded both voltage and current within the electrical circuit. The maximum electrical power (P_{max}) was calculated using:

$$P_{max} = I_{mp} V_{mp} \tag{1}$$

Where I_{mp} is the current at maximum power (A), and V_{mp} is the voltage at maximum power (V).

The electrical efficiency (η), defined as the ratio of useful electrical power output to the total incident solar power, was determined using:

$$\eta_c = \frac{P_{max}}{AG} \tag{2}$$

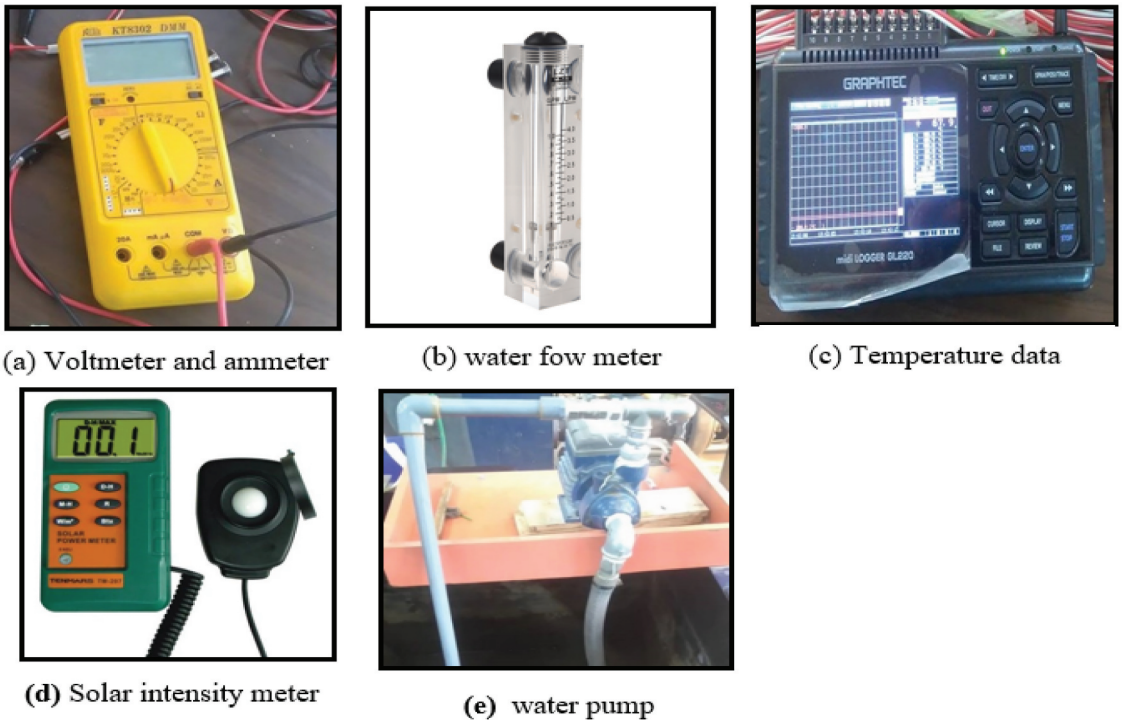


Figure 3. Schematic diagram of the experimental monitoring system showing (a) electrical measurement instruments (digital multimeter), (b) fluid monitoring components (flow meter), and (c) temperature sensors (temperature data logger), (d) solar intensity meter, and (f) water pump with piping circuit.

Where η is the electrical efficiency, A is the surface area of the PV panel (m^2), and G is the solar radiation intensity (W/m^2).

To assess the enhancements from active cooling, two critical performance indicators were analyzed:

• Percentage temperature decrease ($\%\Delta T_{max}$): The maximum temperature decrease due to cooling was calculated using:

$$\%\Delta T_{max} = \left(\frac{T_{without\ cooling} - T_{with\ cooling}}{T_{without\ cooling}}\right) \times 100\% \tag{3}$$

Where: $T_{without\ cooling}$ is peak panel temperature under uncooled operation (°C), and $T_{with\ cooling}$ is peak panel temperature with active cooling (°C).

• Percentage Power Increase (%ΔPmax): The corresponding improvement in electrical output was determined by:

$$\%P_{max} = \frac{P_{with\ cooling} - P_{without\ cooling}}{P_{without\ cooling}} \times 100\% \tag{4}$$

Where: $P_{without\ cooling}$ is maximum power output without cooling (W), and $P_{with\ cooling}$ is maximum power output with cooling (W).

Table 3 provides a comprehensive overview of thermal and structural properties of porous media (flint) integrated into cooling channels. Porous media are materials containing interconnected pores or voids that permit fluid or gas transport through their structure (Mansour et al. 2023). These materials play a vital role in diverse uses, including heat transfer enhancement, fluid flow regulation, and filtration systems (Aljabarin et al. 2022). Within energy and thermal management systems, the selection of porous media significantly influences efficiency, as their permeability, porosity, and thermal conductivity directly affect heat dissipation rates and fluid flow resistance. Their application ranges from passive cooling in PV systems to advanced heat exchangers in concentrated solar power plants.

Porosity, in particular, refers to the ratio of void (empty) space within a solid material to its total volume. It represents the material's capacity to allow the passage or storage of fluids through

Table 3. Properties of the Flint porous material integrated into the cooling system.

Particle Diameter (mm)	Porosity	Total Volume (L)	Fluid Volume (L)	Density (kg/m^3)	Thermal Conductivity (W/m^2. K)	Specific Heat Capacity (J/kg.K)
2	0.35	9.18	5.96	2630	3.5	740
5	0.4	9.18	5.5	2630	3.5	740
10	0.48	9.81	4.77	2630	3.5	740
11	0.5	9.18	4.59	2630	3.5	740

these voids, making it a critical parameter in cooling systems that rely on porous media to facilitate efficient heat exchange. In this study, flint was employed as a natural porous medium within the cooling channel located beneath the PV panel. The spaces between the flint particles form a network of voids that increases the contact area between the coolant (water) and the solid medium. This expanded surface area enhances heat dissipation from the PV panel through both conductive and evaporative mechanisms. To quantify porosity, a two-step experimental method was utilized. First, the total volume of water required to fill the empty cooling channel was recorded. Next, the channel was filled with flint, and water was added again until the channel was completely filled. The difference in water volume between the two steps corresponds to the void volume between the flint particles. The porosity (ϕ) was then calculated using Eq. (5):

$$\phi = 1 - \frac{\forall_f}{\forall_t} \tag{5}$$

where ϕ is the porosity, $\forall_f$ is the volume of fluid added after filling the channel with flint (void volume), and $\forall_t$ is the total volume of the channel.

The porous materials in this study were selected based on key properties, including permeability, pore structure, and thermal conductivity, which critically influence fluid flow and heat transfer. Permeability determines how easily fluid moves through the material, while pore structure affects fluid interaction with internal surfaces, impacting flow distribution and retention. Additionally, thermal conductivity plays a crucial role in heat dissipation, making it particularly significant for applications requiring effective thermal regulation, such as PV panel cooling systems.

2.3. System configuration

The present study quantitatively evaluated multiple cooling approaches to enhance PV system efficiency. Four PV panels of the same size (54 × 34 cm each) were employed in the tests. The cooling system setup, depicted in Figure 2, illustrates the integration of cooling mechanisms with the PV modules. To ensure data reliability, all tests were conducted under carefully controlled environmental conditions, with continuous monitoring of solar irradiance, air velocity, and ambient temperature.

In the first part of the study, the aim was to compare the cooling efficiency of four distinct experimental conditions: cooling with porous media, water cooling alone, a combination of both water cooling and porous media, and a baseline condition with no cooling. This comparison was crucial for understanding the relative effectiveness of each technique in controlling the temperature rise of the PV panels, which directly affects their efficiency and lifespan.

The next phase examined the effect of flint porous media porosity on cooling efficiency. Three PV panels were equipped with cooling channels containing flint porous materials with different porosity sizes (0.35, 0.40, and 0.48). Thermal management performance was assessed at flow rates ranging from 1 to 2 L/min, with each condition tested for 3 days and repeated to ensure consistency, totaling 9 days.

Finally, the study evaluated the impact of volumetric flow rates on PV cooling. Four panels were tested: one operating under ambient conditions (uncooled) and three with 0.35-porosity media at flow rates of 1, 2, and 3 L/min. Each 24-hour trial was repeated over 3 days to ensure reliability, aiming to determine the optimal flow rate for effective cooling and sustainable water circulation.

Table 4. Instrument specifications and measurement uncertainties.

Instrument	Measured parameters	Accuracy	Range	Uncertainty
Solar power meter	G	±1 W/m^2	0–2000 W/m^2	0.57 W/m^2
Ammeter	I	±1 m A	0–10 A	0.5×10^{-3} A
Voltmeter	V	±1 mV	0–200 V	0.5×10^{-3} V
Thermocouple	T	±0.1°C	−210 to 200°C	0.57×10^{-3} °C

2.4. Uncertainty analysis

Uncertainty (U) refers to the degree of variability or potential error in experimental measurements or results. It arises from factors such as limitations in instrumentation, variations in environmental conditions, and imperfections in the experimental setup. Quantifying uncertainty is essential for assessing the reliability and precision of data, ensuring that conclusions drawn from the experiment are robust and valid. To assess the reliability of the experimental results, an uncertainty analysis was conducted in this study using the following equations. The relative uncertainty in the output power (P), area (A), and electrical efficiency (η) was calculated using Eqs. (6) to (8), respectively, as follows (Masalha, Badran, and Alahmer 2025):

$$\frac{UP}{P} = \sqrt{\left(\frac{U_I}{I}\right)^2 + \left(\frac{U_V}{V}\right)^2} \tag{6}$$

$$\frac{UA}{A} = \sqrt{\left(\frac{UW}{W}\right)^2 + \left(\frac{UL}{L}\right)^2} \tag{7}$$

$$\frac{U\eta}{\eta} = \sqrt{\left(\frac{UP}{P}\right)^2 + \left(\frac{UA}{A}\right)^2 + \left(\frac{UG}{Grad}\right)^2} \tag{8}$$

Where w and L represent the cooling channel's width and length, respectively, while G denotes solar radiation intensity. Using Equation (7), the area uncertainty was computed as 0.17%. Similarly, Equation (8) yielded an electrical efficiency uncertainty of 0.2%. Uncertainties in temperature, current, and voltage measurements were derived from instrument specifications, whereas power and efficiency uncertainties were determined through Equations (6)–(8). Table 4 presents the accuracy ranges and uncertainty values for all measurement instruments used in the study.

3. Results and discussion

This study systematically examined the thermal performance of Flint porous media cooling through two primary investigations. First, the investigation proceeded in two phases: initially assessing the thermal regulation impact of porous media with varying porosity levels, then systematically testing multiple flow rates to determine the optimal cooling strategy for each porosity variation.

To comprehensively assess these parameters, we designed three distinct experimental scenarios, each focusing on key aspects of cooling efficiency. These scenarios were designed to investigate crucial factors affecting system performance and heat dissipation. By systematically examining the effects of porosity size and water flow rate under varying operating conditions, the study aimed to enhance the cooling mechanism and determine the most efficient cooling combination for improved thermal regulation.

3.1. Impact of cooling strategies

This scenario systematically evaluated three distinct PV cooling approaches under stable solar irradiance conditions over a seven-hour operational period. The experimental setup compared baseline uncooled panels against two active cooling configurations: conventional water-cooled and advanced flint porous media ($\phi = 0.35$) systems, with coolant flow rates.

Figure 4 illustrates the impact of varying cooling flow rates on PV surface temperature. The thermal analysis revealed substantial differences between cooling methods. Uncooled panels reached peak

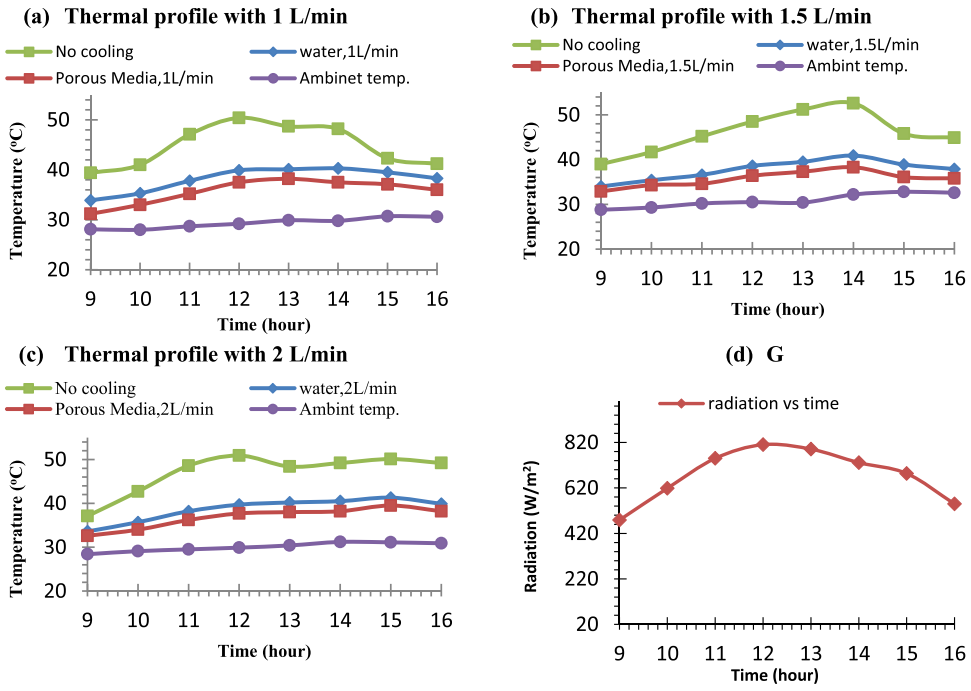


Figure 4. PV panel temperature response to: (a) 1, (b) 1.5, and (c) 2 L/min flow rates, with (d) solar radiation profiles across cooling methods.

temperatures of 50.9°C at midday, highlighting the severe thermal stress experienced by conventional PV systems. Water cooling effectively reduced this peak temperature to 39.7°C at 2 L/min, representing a 22% improvement. The porous media cooling system achieved even better performance, maintaining a peak temperature of just 37.7°C (26% reduction) at the same flow rate. Similarly, at 1 L/min and 1.5 L/min, the uncooled panel reached 48.7°C and 48.5°C, respectively, while the water cooling maintained lower temperatures of 39.6°C and 38.6°C. The porous media-cooled panel further improved cooling performance, reducing temperatures to 38.2°C and 36.4°C, achieving 21% and 24% reductions, respectively. These results align with (Poudel, Zou, and Maroo 2022), which found that porous nanochannels can reduce PV surface temperatures by an average of 31.5°C, demonstrating the potential of advanced porous structures in thermal management. Similarly, Anita and Shruti (Soni and Tiwari 2023) demonstrated that water cooling applied to both sides of a PV panel can reduce surface temperatures by up to 20°C and enhance electrical efficiency by approximately 14.1%.

While the temperature differences across flow rates may appear relatively small in Figure 4, it is important to note that the experiments were conducted on separate days with distinct environmental and operational conditions. The primary aim of this figure is to contrast the performance of the three cooling strategies rather than to isolate the effect of flow rate. To more accurately assess the influence of flow rate under consistent and controlled conditions, a complementary experiment was conducted and presented in Figure 10. In this experiment, all flow rates were tested on the same day under identical environmental settings. Figure 10 thus provides a direct comparison of different flow rates under uniform conditions, revealing clearer distinctions in cooling performance and validating the enhanced thermal management achieved through increased flow in porous media cooling.

This enhanced cooling effect is attributed to the porous material's ability to increase heat dissipation through its large surface area, facilitating efficient heat exchange. The flint porous media's exceptional cooling capacity originates from its unique structural and material properties. The 0.35 porosity creates an extensive internal surface area (~100–1000× greater than smooth surfaces), while its flint composition provides notable thermal conductivity (3.5 W/m·K), as detailed in Table 3.

Figures 5 and 6 illustrate the impact of cooling on PV power generation and electrical efficiency, demonstrating a clear performance enhancement with cooling implementation. The uncooled panel achieved a peak power output of 22.7 W with an electrical efficiency of 13.3%. Water cooling increased

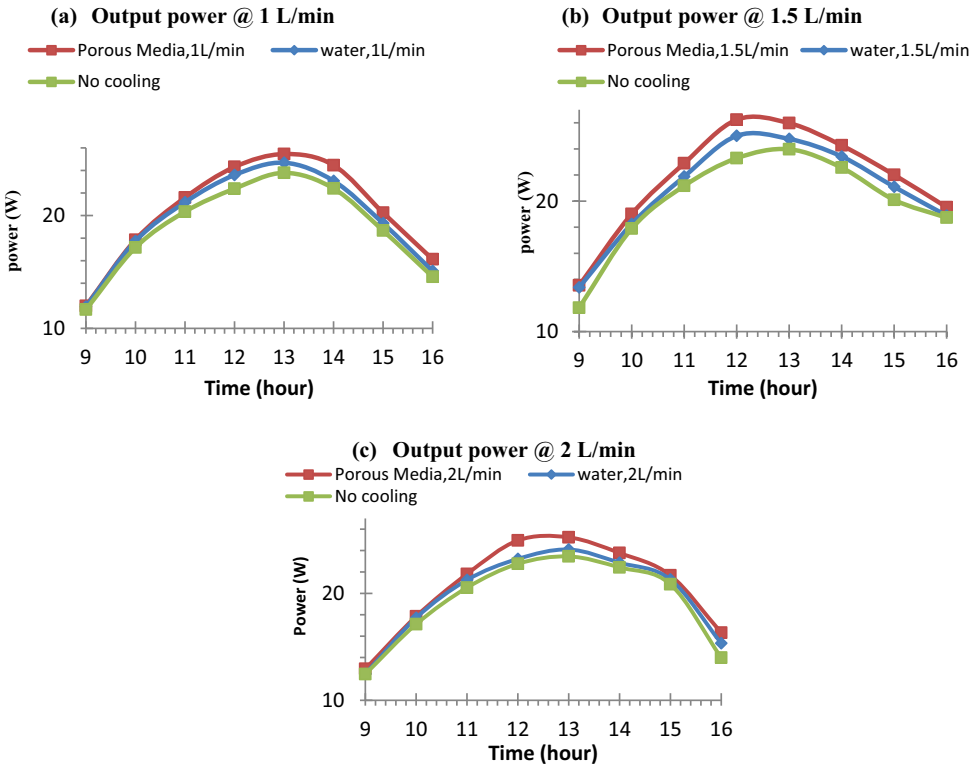


Figure 5. Effect of flow rates on PV output power: (a) 1, (b) 1.5, and (c) 2 L/min, for different cooing strategies.

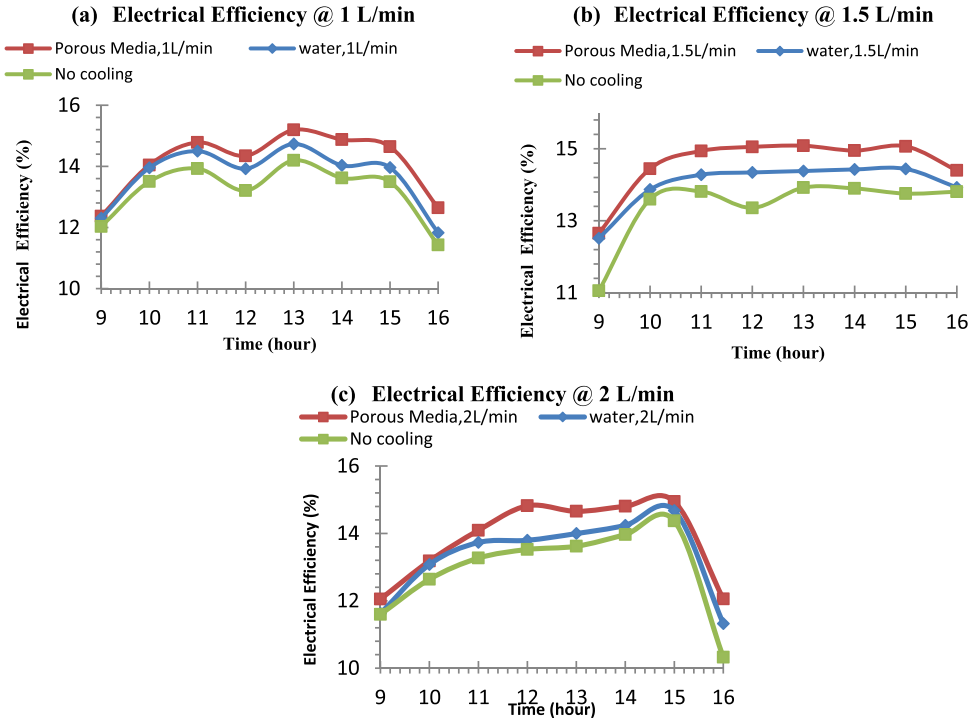


Figure 6. PV electrical efficiency response to: (a) 1, (b) 1.5, and (c) 2 L/min flow rates for different cooling methods.

these values to 24 W and 14.3%, corresponding to a 5.7% increase in power output and a 7.5% boost in electrical efficiency. The porous media cooling method yielded the highest gains, achieving 25.7 W and 15.1% efficiency, representing a 12.3% and 13.5% improvement in both power output and electrical efficiency, respectively, at the optimal 2 L/min flow rate. Lower flow rates still showed benefits, with 7.6%

and 8.7% power improvements at 1 L/min and 1.5 L/min, respectively, when using porous media cooling. The consistent enhancement in both power output and efficiency across all tested conditions confirms that active cooling effectively mitigates the negative impact of temperature rise on PV performance. These findings align with (Wang et al. 2024), and similar advancements have been reported with other innovative cooling approaches. For instance, absorbent hydrogel evaporative cooling combined with 3D porous copper foam has been shown to enhance electrical efficiency by up to 12.84% over natural cooling methods, further highlighting the potential of advanced thermal management techniques in PV systems. Additionally, the integration of porous materials with PCMs has been shown to further enhance cooling performance, reducing PV module temperatures by as much as 21.5°C while simultaneously increasing power output (Firoozzadeh and Shiravi 2022).

The remarkable effectiveness of flint-based porous media in PV cooling systems stems from its unique ability to simultaneously optimize multiple heat transfer mechanisms. The intricate network of interconnected pores creates an expansive internal surface area – typically hundreds of times greater than conventional smooth surfaces. This complex architecture induces microscale turbulence that disrupts thermal boundary layers, while capillary action ensures uniform water distribution across the entire PV surface. The flint material itself (primarily SiO_2) contributes significantly through its favorable thermal properties, with its 0.35 porosity striking an optimal balance between fluid permeability and solid conduction pathways.

Three synergistic cooling mechanisms work within this system: forced convection from the circulating water, evaporative cooling at the numerous air–water interfaces within the pore network, and efficient conductive heat transfer through the solid matrix. This multi-modal approach becomes particularly effective at higher flow rates (2 L/min in experiments), where the porous structure enhances the Nusselt number compared to traditional channel flow. The resulting micro-vortices and extended surface effects boost overall heat transfer coefficients.

3.2. Impact of porosity

This study investigates the effect of flint-based porous media porosity on the cooling efficiency of PV panels. Four PV configurations were compared: three panels with integrated porous media cooling channels (porosity levels: 0.35, 0.45, and 0.48) and one uncooled reference panel for baseline performance. The focus of the investigation includes the thermal regulation capabilities of different porous structures, their impact on PV electrical output, and the identification of the ideal porosity and flow rate combination for optimal cooling performance. The selected porosity range (0.35–0.48) represents a key transition zone where lower porosity (0.35) enhances surface area for heat transfer, higher porosity (0.48) improves fluid permeability, and intermediate values (0.45) strike a balance between both characteristics. To guarantee experimental consistency, all PV panels were tested under standardized environmental conditions with consistent solar radiation levels, ambient temperature, air velocity, and relative humidity. Cooling effectiveness was evaluated at various flow rates (1–2 L/min). The results, shown in Figures 7–9, highlight variations in surface temperature (Figure 7), power output (Figure 8), and electrical efficiency (Figure 9) across different porosity levels and flow rates. At the optimal 2 L/min flow rate, the 0.35 porosity configuration achieved a remarkable 27% temperature reduction, and 12% power increase compared to uncooled operation, outperforming higher porosity alternatives. This enhanced performance stems from the intricate interplay between pore structure and heat transfer mechanisms – the reduced pore size in 0.35 porosity media creates greater surface area for heat exchange while maintaining adequate fluid permeability. The results demonstrate a clear porosity-dependent trend in cooling effectiveness, with temperature reductions scaling from 22% at 0.48 porosity to 27% at 0.35 porosity under maximum flow conditions. Correspondingly, power output improvements ranged from 6.7% to 12% across the porosity spectrum, confirming that pore structure optimization represents a critical factor in PV cooling system design.

The superior cooling performance observed with the 0.35 porosity flint media at a 2 L/min flow rate stems from an optimal combination of thermal and hydrodynamic factors. The lower porosity structure creates a denser network of smaller pores, significantly increasing the available surface area for heat transfer compared to higher porosity configurations. This enhanced surface area facilitates more efficient conductive heat transfer through the solid flint matrix while simultaneously improving convective heat exchange at the fluid-solid interfaces. The constrained pore

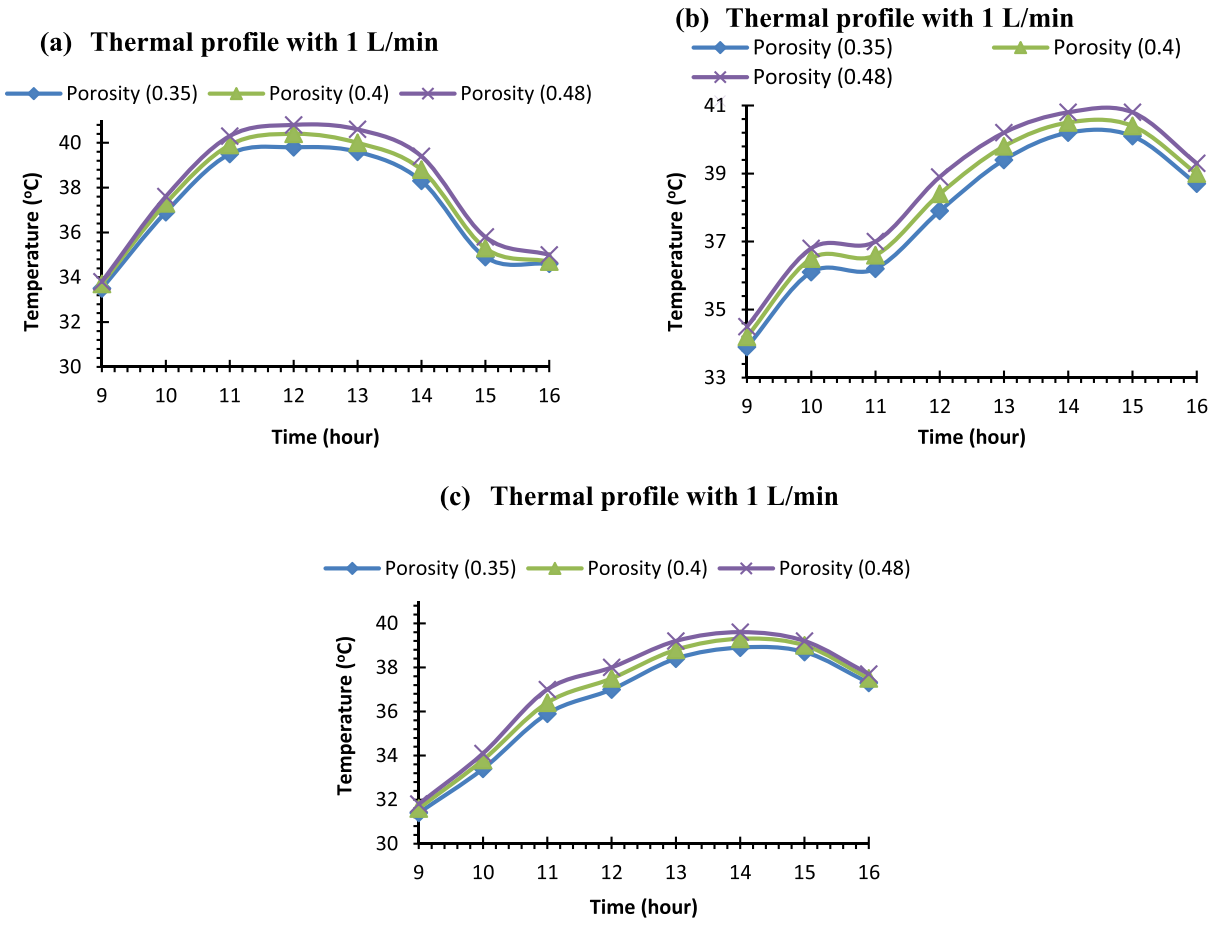


Figure 7. Porosity-dependent PV temperature profiles at: (a) 1, (b) 1.5, and (c) 2 L/min flow rate.

channels generate beneficial microscale turbulence that disrupts thermal boundary layers and promotes thorough coolant mixing, boosting the effective heat transfer coefficient (Masalha, Badran, and Alahmer 2025).

While lower porosity values (< 0.35) might theoretically offer greater surface area for heat transfer, several practical limitations make them less suitable for PV cooling applications. First, excessively low porosity significantly increases flow resistance, requiring substantially higher pumping power that could offset the energy gains from improved cooling. The pressure drop across the porous media would increase exponentially below 0.35 porosity, making the system energetically inefficient. Second, very dense porous structures are more prone to clogging from mineral deposits or particulates in the coolant, compromising long-term reliability. Third, below 0.35 porosity, the reduced pore connectivity begins to create uneven coolant distribution, leading to localized hot spots on the PV surface. Additionally, manufacturing constraints make the consistent production of high-quality porous media difficult below this porosity threshold while maintaining structural integrity.

3.3. Flow rate effects

This study examines the influence of coolant flow rate on PV panel temperature regulation using flint porous media cooling channels. As shown in Figure 10, three PV panels equipped with 0.35 porosity flint-based cooling systems were compared against an uncooled reference panel under identical environmental conditions. The controlled experiments evaluated three flow rates (1, 2, and 3 L/min), with Figure 10 clearly demonstrating the progressive temperature reduction achieved of 21%, 23%, and 26% for each increment, respectively.

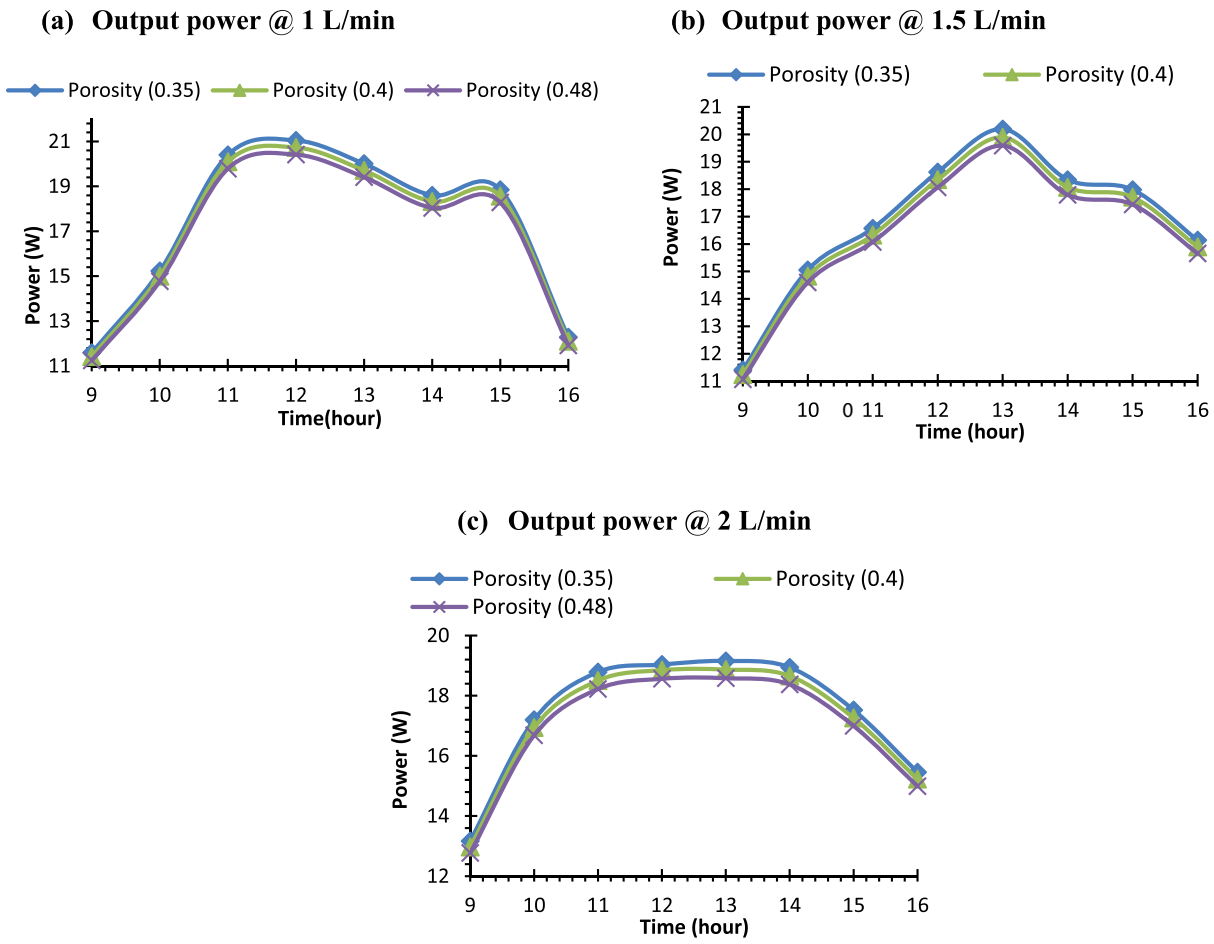


Figure 8. Porosity-dependent power output at: (a) 1, (b) 1.5, and (c) 2 L/min flow rates.

The cooling enhancement visible in Figure 10 results from several flow-dependent mechanisms. Higher flow rates reduce thermal boundary layer thickness while increasing coolant refresh frequency at heat exchange surfaces. Particularly at 3 L/min, the system achieves turbulent flow conditions that generate micro-vortices, significantly improving heat transfer efficiency. While 3 L/min shows best thermal regulation (26% reduction), practical implementation requires balancing these benefits against pumping power requirements.

The synergistic combination of 0.35 porosity with a 2 L/min flow rate represents a significant technological advancement in PV cooling solutions. This configuration delivers superior thermal regulation while maintaining practical implementation feasibility. The fundamental heat transfer mechanisms underlying these results can be quantitatively explained through several critical relationships:

- The convective heat flux (q”$_{conv}$) is governed by:

$$q''_{conv} = h(T_s - T_\infty) \tag{9}$$

where the heat transfer coefficient (h) directly determines cooling effectiveness, Ts is a surface temperature (°C), and T_∞.is ambient temperature (°C). The results confirm that the 0.35 porosity maximizes this coefficient through enhanced surface interactions.

- The convective heat transfer coefficient ($h_{c,f}$) is a function of the Nusselt number, as given by:

$$h_{c,f} = \left[\frac{D_{hd}}{Nu_{pm-f}.k_e}\right]^{-1} \tag{10}$$

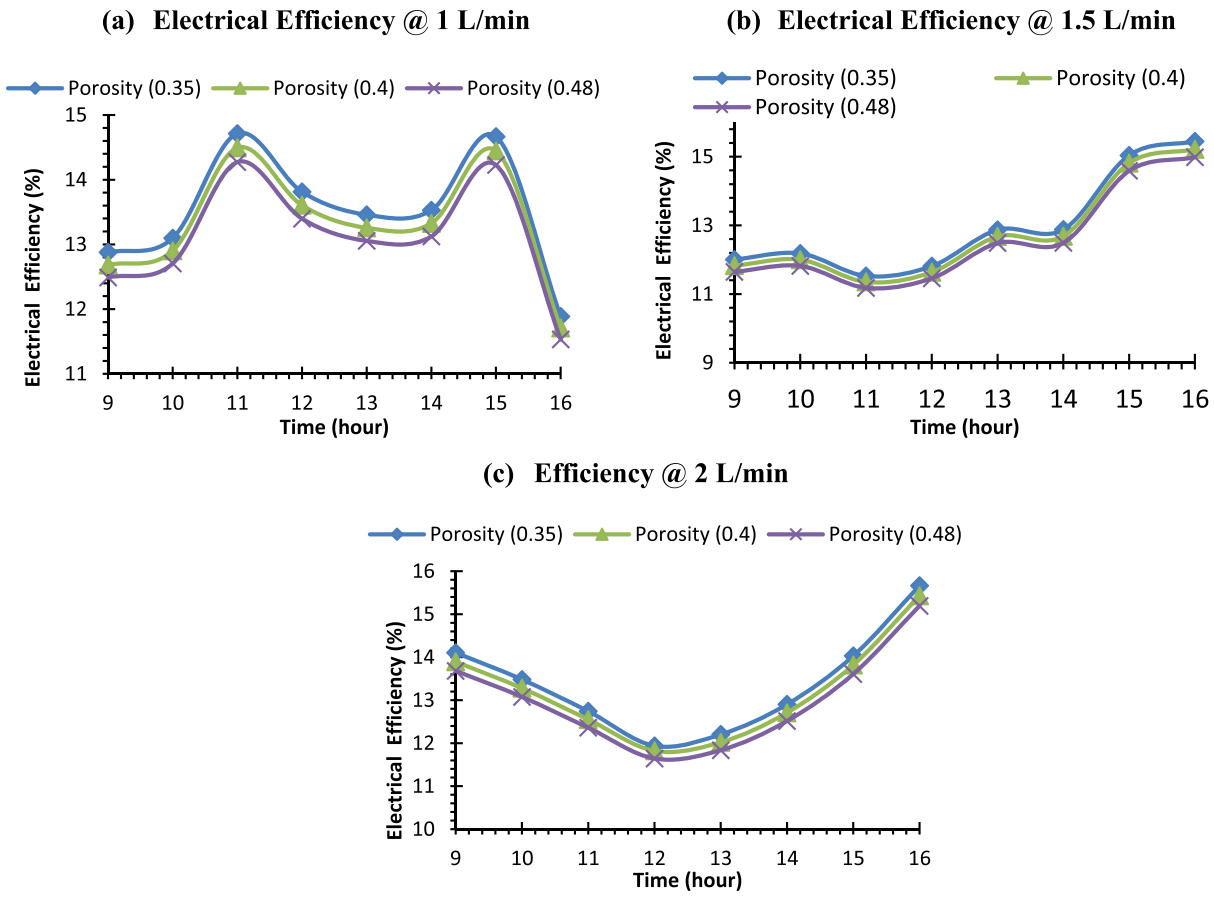


Figure 9. Electrical efficiency dependence on porosity size at: (a) 1, (b) 1.5, and (c) 2 L/min water flow rates.

Where, D_{hd} is the hydraulic diameter (m), Nu_{pm-f} is the Nusselt number at the porous medium – fluid interface, k_e is the effective thermal conductivity (W/m·K). This equation demonstrates how porous media properties influence thermal performance.

• The Nusselt number for porous media is expressed as:

$$Nu_{pm-f} = \left(\frac{0.255}{\phi}\right) Re_{pm}^{\frac{2}{3}}.Pr_f^{1/3} \tag{11}$$

Where, ϕ is porosity of the porous medium, Re_{pm} is the Reynolds number in the porous medium, Pr_f is the Prandtl number of the fluid. This relationship quantitatively explains our experimental observation that reduced porosity ($\phi = 0.35$) yields higher Nusselt numbers and consequently better cooling performance.

The flow rate analysis reveals that increased coolant velocity produces two beneficial effects: Elevated Re_{pm} by enhancing turbulent mixing and improved convective heat transfer coefficients.

The cooling water circulated at 2–3 L/min in the hybrid PV/T system represents a valuable thermal resource that can be effectively repurposed to enhance overall system sustainability. After absorbing waste heat from PV modules (typically reaching 30–40°C), the warmed water can be used in several agricultural and aquacultural applications. For example, in greenhouse environments, this thermally enhanced irrigation water supports optimal root-zone temperatures and reduces the demand for freshwater. Çakir et al. (Çakir et al. 2017) demonstrated that such systems can significantly enhance crop growth rates compared to traditional irrigation methods. Similarly, in aquaculture, the moderately heated water can help maintain stable pond temperatures, benefiting fish farming operations (Mugwanya et al. 2022).

This approach is particularly advantageous in arid regions, where it simultaneously improves PV efficiency and supports water conservation efforts. The system is designed with a closed-loop configuration and insulated water storage to minimize thermal losses and allow repeated reuse of water, with only minimal makeup water needed to offset evaporation. In this way, the cooling function becomes more

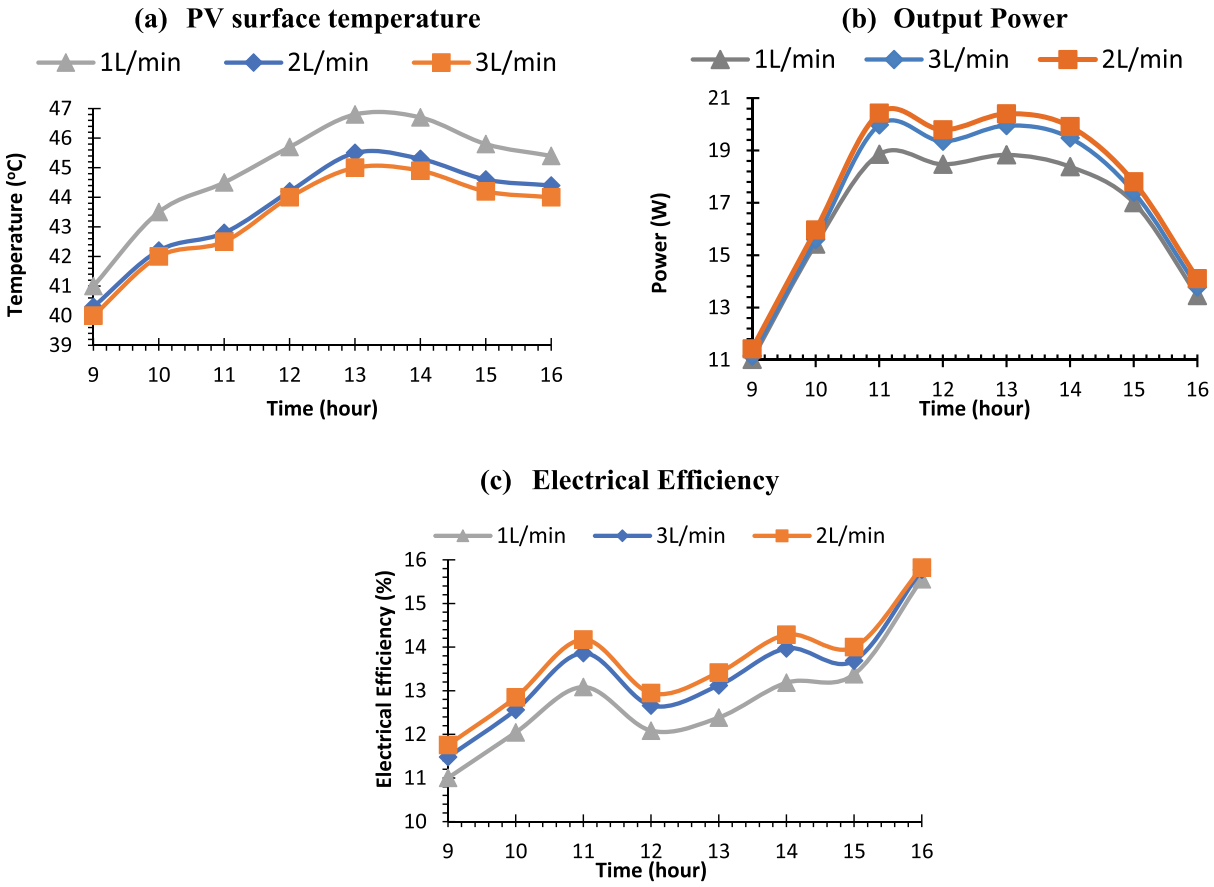


Figure 10. Performance metrics for porosity 0.35: (a) PV temperature, (b) power output, and (c) electrical efficiency versus flow rate.

than just an energy expense – it is transformed into a valuable input for agricultural sustainability, fostering a synergistic integration between renewable energy production and food systems.

4. Economic assessment

Before evaluating the economic viability of different cooling systems for PV panels, a Monte Carlo simulation with 1,000 iterations was conducted using collected production data as a baseline. The simulation incorporated a ± 20% variation in annual energy output to account for operational and environmental uncertainties. The levelized cost of energy (LCOE) for each cooling system was then calculated based on the Monte Carlo simulation outputs, ensuring a fair comparison that accounts for performance variability and uncertainty (Awad et al. 2020).

In this study, the calculation of the LCOE for the PV with and without the cooling systems was based on several key assumptions that account for technological, operational, and financial factors affecting the overall cost of energy production. The LCOE is calculated using Eq. (12) (Al-Jaraden, Ayadi, and Alahmer 2023):

$$LCOE = \frac{i + \sum_{t=0}^{T} \frac{O_t}{(1+r)^t}}{\sum_{t=0}^{T} \frac{S_t}{(1-d)^t}} \tag{12}$$

Where T is the system life, t is the time period in years, i is the initial investment, O_t is the operation and maintenance costs, r is the discount rate, S_t is the rated energy output, and d is the annual module degradation factor. The following assumptions and cost components are used to calculate the LCOE, as shown in Table 5.

Table 5. Assumptions and cost components for LCOE calculation.

Parameter	Value	Description
Annual degradation	0.50%	Expected yearly performance degradation
O&M escalator (%)	3.00%	Annual escalation rate of O&M costs
Discount rate (%)	3%	Discount rate
system life (years)	15	Expected operational lifetime of the system
operating days per year	300	Assumed annual operational days
O&M cost ($/w)	0.047	Operation and maintenance cost per watt
Water cost ($/w)	0.009	Water-related cost per watt (included in O&M expenses)
Pump cost ($/w)	0.058	Cost per watt for the pumping system (included in initial capital investment)
Flint cost ($/w)	0.023	Cost per watt associated with Flint (included in initial capital investment)
Storage tank + pipes system ($/w)	0.6	Cost per watt for the storage tank and piping system (initial investment)

Table 6. LCOE for various PV cooling configurations.

System Type	LCOE ($/kWh)
PV (no cooling)	$0.073
PV + Water Cooling	$0.105
PV + Water Cooling + Flint	$0.103

The Monte Carlo simulation results reveal that a PV system without cooling yields an estimated annual energy output of 45,837 Wh. Incorporating water cooling increases the output to approximately 47,534 Wh per year. When both water cooling and a flint coating are applied, the annual energy production rises further to around 49,222 Wh. The Levelized Cost of Energy (LCOE) comparison for these three configurations is presented in Table 6. The LCOE analysis reveals that the uncooled PV system exhibits the lowest LCOE at $0.073/kWh due to the absence of additional cooling components. However, this configuration also delivers the lowest annual energy output due to higher operating temperatures, which negatively impact efficiency. The inclusion of water cooling increases the LCOE to $0.105/kWh, primarily because of added capital and operational costs such as pumping and water consumption. Despite this increase, the water-cooled system benefits from improved energy production, which partially offsets the additional costs. The hybrid system combining water cooling with flint porous media shows a slightly lower LCOE of $0.103/kWh compared to water cooling alone. This reduction is attributed to the enhanced thermal management provided by the flint, which improves overall energy output and system performance, thus achieving a better balance between cost and efficiency. These results highlight the trade-offs between upfront and operational expenses versus energy yield improvements when implementing cooling technologies in PV systems. It should be highlighted that the use of a cooling system helps eliminate hot spots on the PV panel surface, thereby extending the panel's lifetime, which also contributes to long-term economic benefits (Hadipour, Rajabi Zargarabadi, and Rashidi 2021).

Following the Monte Carlo simulation, a sensitivity analysis was conducted to evaluate the effect of annual degradation rates on the LCOE for various PV cooling configurations, as summarized in Table 7. The analysis evaluated degradation rates ranging from 0.3% to 0.7% per year. Results showed that as degradation increased, the LCOE increased slightly across all systems. For the uncooled PV system, LCOE increased from $0.072/kWh at 0.3% degradation to $0.074/kWh at 0.7%. In comparison, the PV system with water cooling showed an increase from $0.104/kWh to $0.107/kWh, while the PV system with both water cooling and flint coating increased from $0.101/kWh to $0.104/kWh. These findings highlight the importance of long-term performance stability in economic evaluations of PV cooling technologies.

Table 7. Effect of annual degradation on LCOE for uncooled, water-cooled, and flint-enhanced water-cooled PV systems.

Annual degradation	Uncooled PV	PV with water cooling	PV system with integrated water cooling and porous flint media
0.3%	$0.072	$0.104	$0.101
0.4%	$0.072	$0.105	$0.102
0.5%	$0.073	$0.105	$0.103
0.6%	$0.073	$0.106	$0.103
0.7%	$0.074	$0.107	$0.104

Table 8. Comparative performance and environmental impact metrics of 30 W polycrystalline PV modules under three cooling configurations: uncooled, water-cooled, and hybrid porous media-water cooling.

Metric	Calculation method	Uncooled PV	Water-cooled PV	PV with integrated water cooling and porous material
Annual energy generation (kWh/year)	–	45.84	47.53	49.22
Annual CO_2 emissions avoided (kg CO_2/year)	Annual Energy × Grid emission factor (0.5 kg CO_2/kWh)	22.92	23.77	24.61
Embodied energy (kWh)	Installed Capacity (0.03 kW) × Embodied energy (3,000 kWh/kW)	90	90	90
Embodied CO_2 emissions (kg CO_2)	Embodied energy × Emission factor (100 g CO_2/kWh)	9	9	9
Net CO_2 mitigation over 15 years (kg CO_2)	(Annual CO_2 Avoided × 15) − embodied CO_2 emissions	334.8	347.55	360.15
Annual carbon credit value ($/year)	(Annual CO_2 avoided ÷ 1,000) × $14.5	$0.332	$0.345	$0.357

5. Environmental assessment

Renewable energy technologies, particularly PV systems, are instrumental in reducing greenhouse gas emissions and addressing climate change. This assessment examines the environmental performance of a small-scale system based on 30 W polycrystalline solar modules. The analysis includes key indicators such as annual energy output, avoided CO_2 emissions, embodied energy and emissions, net CO_2 mitigation over a 15-year operational life, and potential carbon credit revenues. A carbon credit price of $14.5 per metric ton of CO_2 equivalent (tCO_2e) is used for the calculations (Badran et al. 2025).

Key assumptions in this analysis include:

- Panel capacity: 30 Wp (0.030 kWp)
- Embodied energy: 3,000 kWh per kWp
- Embodied CO_2 emissions: 100 g CO_2 per kWh
- System lifetime: 15 years
- Grid emission factor: 0.5 kg CO_2 per kWh (global average)
- Carbon credit value: $14.5 per metric ton of CO_2 equivalent (tCO_2e)

Table 8 presents a comparative analysis of the performance and environmental impact of 30 W polycrystalline PV modules under three cooling configurations: uncooled, water-cooled, and a hybrid system integrating water cooling with porous material. The data indicate a clear improvement in annual energy generation with the addition of cooling mechanisms, with the hybrid system yielding the highest output at 49.22 kWh/year. Consequently, the annual CO_2 emissions avoided increase slightly across the configurations, reflecting improved environmental performance. The net CO_2 mitigation over a 15-year period shows incremental gains, reaching up to 360.15 kg CO_2 for the hybrid cooling system. Financially, the annual carbon credit gain is modest (around $0.33–$0.36/year) due to the small system size, but scaling up capacity would significantly improve both environmental and economic outcomes.

6. Comparative analysis: current vs. literature studies

Table 9 compares various cooling methods for PV panels, evaluating their effectiveness in reducing surface temperature, increasing power output, and improving electrical efficiency. The results show significant variation across techniques, with water-based cooling (spray, hybrid air-water, and double-sided glazing) achieving temperature reductions of 7–24°C and efficiency gains up to 14%. The current study's flint porous media performs competitively, reducing temperature by 22°C while increasing efficiency by 13.5%, positioning it as a promising alternative to water-intensive methods. However, inconsistencies in reported metrics (e.g., absolute vs. percentage reductions) and missing data in some studies highlight the need for standardized testing to enable fair comparisons.

Table 9. Comparison of PV panel cooling techniques.

Source	Cooling Approach	Temperature Drop (°C)	Power Increase (%)	Electrical Efficiency Gain (%)
(Basem et al. 2024)	Water-spray cooling	41°C (final temperature)	Not specified	3.9%
(Ali, Abdalrahman, and Wahid 2019)	Water spray	7°C	2.5%	Not specified
(Mohammed, Mohammed, and Sanad 2019)	Double-sides water glazing chambers	50.06% reduction	12.69%	14.2%
(Nabil and Mansour 2022)	Water front cooling	29.37% reduction	7.43%	6.84%
(Abdel-Raheim Amr et al. 2024)	Evaporative cooling	44.8% reduction	Not specified	1.8% (continuous) 1.7% (intermittent)
(Alktranee and Bencs 2023)	cotton wicks integrated with fins (passive cooling)	31.4% reduction	66.6%	From 3.12% to 8.6%
(Soni and Tiwari 2023)	Water cooling (both sides)	20°C reduction	Not specified	14.1%
Current Study	Flint Porous media	22°C reduction	12.3%	13.5%

7. Conclusions

This study investigated the effectiveness of flint-based porous media in enhancing PV cooling performance through integrated liquid cooling systems. The study systematically evaluated the synergistic effects of integrated water-cooling systems with varying porous media configurations, focusing on two critical parameters: porosity characteristics (0.35–0.48) and coolant flow rates (1–3 L/min). The study compared uncooled PV panels with those cooled using water alone, porous media alone, and hybrid porous media-liquid cooling configurations. The key findings are summarized as follows:

- The flint porous media (porosity 0.35) reduced PV surface temperatures by 26% compared to uncooled panels, outperforming conventional water cooling (22% reduction).
- The porous media cooling system improved PV electrical efficiency by 13.5% and power output by 12.3% at optimal flow rates.
- A lower porosity of 0.35 provided the most effective cooling performance due to its increased surface area for heat transfer, outperforming the higher porosity levels of 0.40 and 0.48. This specific porosity offered an optimal balance between surface area for heat exchange and fluid permeability.
- A flow rate of 2 L/min achieved the best balance between cooling efficiency and energy consumption, reducing temperatures by 23–26%.
- The porous media enhanced cooling through forced convection, conduction, and micro-vortex-induced turbulence, maximizing heat dissipation.
- The hybrid cooling system maintained lower peak temperatures (37.7°C vs. 50.9°C for uncooled panels), which improves PV longevity.
- Porous media cooling outperformed water-only cooling by 3–4% in temperature reduction and 5–7% in power output.
- Increasing flow rates from 1 to 3 L/min improved cooling, but 2 L/min was most energy-efficient, avoiding excessive pumping power.
- The flint's thermal conductivity (3.5 W/m·K) and pore structure facilitated efficient heat exchange with minimal pressure drop.
- Monte Carlo-based economic analysis (1,000 iterations) confirmed that the integrated cooling system with porous media achieved a favorable LCOE of $0.103/kWh – lower than water cooling ($0.105/kWh) and offering better performance than the uncooled system despite its lower LCOE ($0.073/kWh).
- The integrated cooling system with porous material increased annual CO_2 avoidance by 3.5% over the uncooled system, totaling 360.15 kg CO_2 mitigation over 15 years. While small-scale carbon credit gains are modest ($0.36/year), the system's scalability offers sizable long-term environmental benefits and supports sustainable PV performance.

One significant limitation of the current experimental setup is the lack of recorded outlet water temperature data. While the surface temperature of the PV modules was measured to evaluate cooling performance, the

outlet water temperature is essential for determining the actual thermal energy extracted from the PV module.

These findings demonstrate that integrating flint porous media with liquid cooling significantly enhances PV performance, providing a viable pathway for improving solar energy efficiency in high-temperature environments. Future work should explore long-term durability and the potential of alternative porous materials for further optimization.

Author contributions

CRediT: **Ismail Masalha:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing; **Omar Badran:** Formal analysis, Investigation, Writing – original draft; **Siti Ujila Masuri:** Conceptualization, Investigation, Methodology; **Razan Alkhazaleh:** Formal analysis, Investigation, Methodology, Software, Writing – review & editing; **Abdel Salam Alsabagh:** Investigation, Writing – original draft; **Ali Alahmer:** Conceptualization, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

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Data availability statement

No data was used for the research described in the article.

Nomenclature

Symbol/Abbreviation	Definition
A	Collector or panel area (m^2)
Ag – MgO	Silver – magnesium oxide
Al_2O_3	Aluminum oxide
d	Annual module degradation factor
D_h(d)	Hydraulic diameter (m)
G	Solar radiation intensity (W/m^2)
H	Hardness (Mohs scale)
h	Convective heat transfer coefficient ($W/m^2 \cdot K$)
i	Initial investment
k	Thermal conductivity (W/m·K)
k_e	Effective thermal conductivity of the porous fluid composite (W/m·K)
L	Cooling channel length (m)
Nu	Nusselt number
Nu_{pm-f}	Nusselt number at the porous medium – fluid interface
P	Power output (W)
PCM	Phase change material
PHP	Pulsating heat pipes
PVC	Polyvinyl chloride
PV	Photovoltaic
PV/T	Photovoltaic – thermal
q''_{conv}	Convective heat flux (W/m^2)
O_t	Operation and maintenance costs
Pr_f	Prandtl number of the fluid
$P_{with\ cooling}$	Maximum power output with cooling (W).
$P_{without\ cooling}$	Maximum power output without cooling (W),
r	Discount rate

Re	Reynolds number
Re_{pm}	Reynolds number in the porous medium
SiC	Silicon carbide
S_t	Rated energy output
STC	Standard test conditions
t	Time period in years
T	Panel surface temperature (°C), or system life in Eq. (12)
$T_{with\ cooling}$	Peak panel temperature under cooled operation (°C)
$T_{without\ cooling}$	Peak panel temperature under uncooled operation (°C)
$T\infty$	Ambient (free-stream) temperature (°C)
T_s	Surface temperature (°C)
TiO_2	Titanium dioxide
U	Uncertainty
Voc	Open-circuit voltage
w	Cooling channel width (m)
η	Electrical efficiency
ϕ	Porosity (fractional void volume)
$\%\Delta P_{max}$	Percentage power increase
$\%\Delta T_{max}$	Percentage temperature decrease
$\forall_f$	Volume of fluid added after filling the channel with flint (void volume)
$\forall_t$	Total volume of the channel

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